The Human Premium and the Anti-AI Utopia: Contrarian Portfolio Construction and Performance Comparison amid the AI Boom

Financial Data Analysis: Final Report

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Translator's note: this English edition follows the original Chinese report. Three figures in the original text were corrected to match the report's own tables and code: the GLD–SPY and TLT–SPY correlations (Section 2.4), the annualized standard deviation of GLD (Section 2.3.3), and the transaction-cost rate (Section 3.2).

1. Research Motivation and Data

1.1 Background and Motivation

Since generative artificial intelligence made significant breakthroughs in late 2022, global capital markets have undergone a structural shift. Capital has concentrated heavily in a small number of large technology companies and the AI-infrastructure supply chain, driving broad indices such as the Nasdaq-100 and the S&P 500 higher. Yet while technology has lifted the market as a whole, asset pricing across industries has diverged sharply. Industries whose business models are easily replaced by AI face valuation re-ratings and selling pressure, whereas industries that possess proprietary intellectual property, physical on-site experiences or information-security defenses, none of which software algorithms can fully replicate, have defensive characteristics whose pricing efficiency in the early stage of technology diffusion has yet to be tested.

Against this backdrop, the core motivation of this study is to examine how well traditional portfolio-optimization models adapt when capital markets face a major technology shock and asset returns are non-normally distributed. The study builds a research universe that includes both "high-risk AI-substitution assets" and "assets with physical defensive characteristics", and applies the optimization framework of Markowitz's Modern Portfolio Theory (MPT), covering the Global Minimum Variance Portfolio (GMVP), the Minimum Variance Portfolio with a target return (MVP) and the tangency portfolio.

Through an out-of-sample rolling backtest, this study tests whether an algorithm driven purely by the historical covariance matrix can automatically achieve dynamic risk avoidance across assets without any ability to forecast future returns. In addition, because the historical covariance matrix is prone to severe estimation error in highly volatile environments, the study also examines the empirical benefit of practical constraints such as a no-short-sale rule in suppressing the "error maximization" problem and guarding against tail risk.

1.2 Asset Classification and Data Sources

To ensure that the covariance matrix captures the heterogeneous responses of different industries to the AI shock, this study selected 37 globally representative stocks and ETFs and divided them into five core groups. Data are daily adjusted closing prices from Yahoo Finance, which remove the effects of dividends and stock splits on return calculations.

The 37 assets are grouped as follows, with the financial rationale for each group:

1. **AI-substitution casualties.** The core business models of this group rely heavily on transmitting entry-level knowledge, writing routine code or basic outsourced services, making them high-risk assets that generative AI can replace directly. When AI makes a breakthrough, this group is expected to bear significant downside risk and valuation re-rating.
    - CHGG (Chegg): online education and homework help
    - FVRR (Fiverr): freelance outsourcing platform
    - UPWK (Upwork): freelance outsourcing platform
    - COUR (Coursera): online course platform

- DLB (Dolby Laboratories): audio and video processing technology (basic audio processing faces competition from AI generation)
- PSO (Pearson): traditional educational publishing and assessment
- LOPE (Grand Canyon Education): education services provider

2. **Intellectual property and moats.** This group owns large, exclusive bodies of content and IP that are strictly protected by law. At a time when AI models need vast amounts of high-quality, lawfully licensed data for training, these companies have strong bargaining power over data licensing and a physical moat that algorithms cannot scrape for free.
   - DIS (Walt Disney): film, entertainment and theme parks
   - NTDOY (Nintendo): closed gaming ecosystem and classic IP
   - NYT (New York Times): media with high-quality news copyrights
   - SONY (Sony Group): entertainment empire spanning music, film and games
   - WMG (Warner Music Group): one of the world's three major record labels
   - CMCSA (Comcast): telecom giant with a strong film library (NBCUniversal) and a broadband moat

3. **Deepfake defenders.** As AI-driven deepfakes, automated social engineering and cyberattacks surge, this group represents the "pick-and-shovel" beneficiaries of new, essential demand created by AI threats, and has the characteristics of a hedge against AI's negative externalities.
   - CRWD (CrowdStrike): endpoint security and threat intelligence
   - PANW (Palo Alto Networks): enterprise network-security platform
   - OKTA (Okta): zero-trust identity and access management
   - EFX (Equifax): credit ratings and identity verification
   - FICO (Fair Isaac): credit-scoring models and fraud-prevention analytics
   - FTNT (Fortinet): network-security firewall equipment

4. **The human experience economy.** The business value of this group rests on "five-sense experiences", in-person social interaction and physical spatial consumption that cannot be digitized or simulated by algorithms. In a highly virtualized AI era, genuine physical experiences should command a higher "human premium".
   - LYV (Live Nation): live concerts and on-site entertainment
   - LVMUY (LVMH): top-tier physical luxury experiences and brand value
   - ABNB (Airbnb): physical lodging and distinctive local experiences
   - MAR (Marriott International): global hotel chain
   - SBUX (Starbucks): the physical "third place" social and coffee experience
   - LVS (Las Vegas Sands): physical gaming and resort experiences
   - DAL (Delta Air Lines): physical international air transport
   - HCA (HCA Healthcare): physical healthcare system
   - BKNG (Booking Holdings): online booking platform for physical travel

5. **Market and benchmark reserves.** To give the optimization models enough room to diversify risk, this study includes a control group covering broad equity indices, bonds and commodities, used to test how

the quantitative models allocate weight to "traditional safe-haven assets" under extreme market conditions.

- SPY (SPDR S&P 500 ETF): US equity benchmark
- QQQ (Invesco QQQ Trust): technology-heavy benchmark
- GLD (SPDR Gold Shares): traditional safe-haven asset (gold)
- TLT (iShares 20+ Year Treasury Bond ETF): traditional safe haven and risk-free substitute (US Treasuries)
- USO (United States Oil Fund): physical crude-oil energy asset
- DBA (Invesco DB Agriculture Fund): physical agricultural commodities and inflation hedge
- VNQ (Vanguard Real Estate Index Fund): physical real estate investment trusts
- EWW (iShares MSCI Mexico ETF): beneficiary of nearshoring and physical supply-chain relocation
- XLU (Utilities Select Sector SPDR Fund): utilities supplying the underlying energy for AI computing (power and water infrastructure)

1.3 Sample Period and Data Processing

This section describes the data-preprocessing procedures required for the empirical study, ensuring the correctness of the subsequent matrix computations and portfolio optimization. The steps are as follows:

1. **Sample period.** The full empirical sample runs from 1 December 2021 to 31 December 2024. After removing holidays and missing values, it covers 774 trading days. To fit the dynamic rolling-window framework, the full period is strictly divided into two parts: the first period, 1 December 2021 to 30 November 2022 (252 trading days), serves as the initial parameter-estimation window; the subsequent period, 1 December 2022 to 31 December 2024 (actual trading through 30 December, 522 trading days), is the out-of-sample testing period. This three-year span covers both the Federal Reserve's rate-hiking turbulence and the generative-AI boom, allowing a full examination of each model's dynamic risk avoidance in a non-normal market.
2. **Return calculation and missing values.** The study uses discrete daily returns as model inputs. Because the assets include cross-listed ADRs (such as Nintendo) and various ETFs, differences in national holidays can misalign the data. All trading days containing missing values are removed with `na.omit()` to keep the data matrix complete and to avoid a non-invertible covariance matrix in later calculations.
3. **Risk-free rate.** When computing Sharpe ratios and the tangency portfolio, the annualized risk-free rate is set at 2.5%. Because the sample period spans both low-rate and rate-hiking phases, the historical average of the 10-year US Treasury yield over the past decade (about 2.5%) is used as the benchmark for a long-term investor's opportunity cost.

2. Individual Asset Characteristics and Descriptive Statistics

2.1 Daily Returns and Cumulative Wealth of Individual Assets

Before optimizing portfolio allocations, the historical price behavior of each asset in the universe must first be analyzed as a time series. Following the course guidelines, this study plots the daily returns and cumulative wealth paths of the 37 assets over the sample period (1 December 2021 to 31 December 2024) to examine their cross-sectional characteristics and dynamic paths.

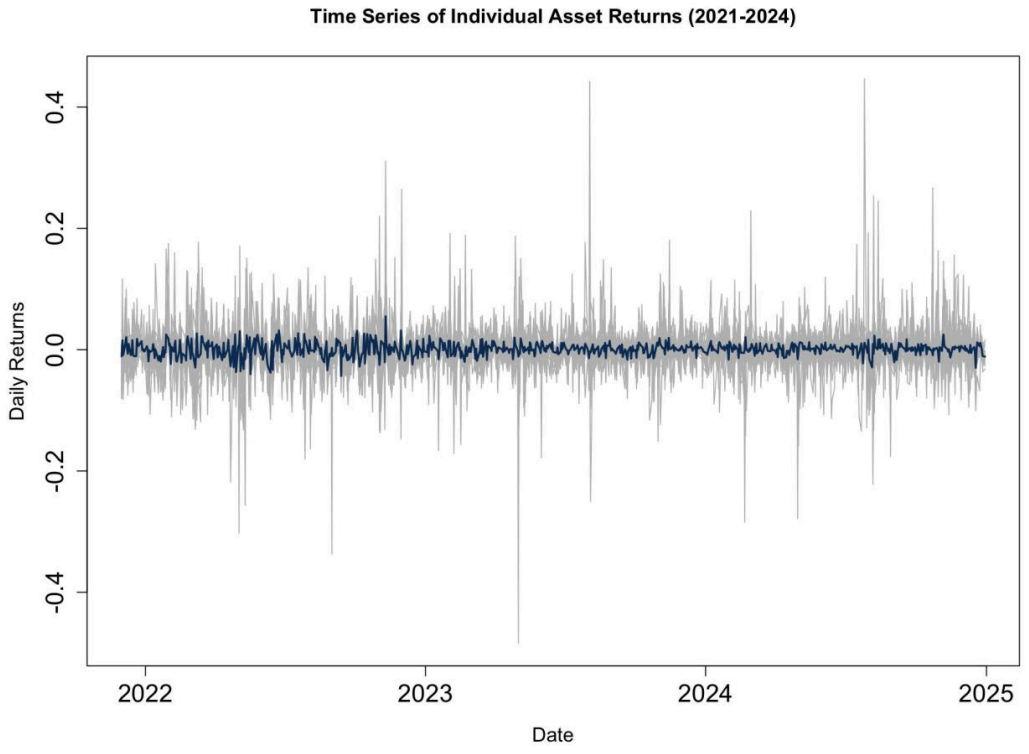


Figure 1: Time series of individual asset daily returns

As Figure 1 shows, all assets display mean reversion, oscillating around zero. However, the figure reveals a striking contrast between the broad market benchmark and individual-stock heterogeneity. SPY (dark blue line), representing the broad market, has very contained fluctuations, with daily returns staying within a ±0.05 band. In contrast, individual assets (gray lines) frequently show extreme spikes during the sample period. For example, in early 2023 one asset suffered a single-day crash approaching −0.5 (a left-tail event), and in mid-2023 and the second half of 2024 there were several single-day surges above 0.4 (right-tail bursts). This highly concentrated volatility clustering and fat-tailed distribution reflects the pricing fragility of individual companies in the face of macro-level high interest rates and the AI technology shock, further confirming the need to eliminate idiosyncratic risk through asset allocation.

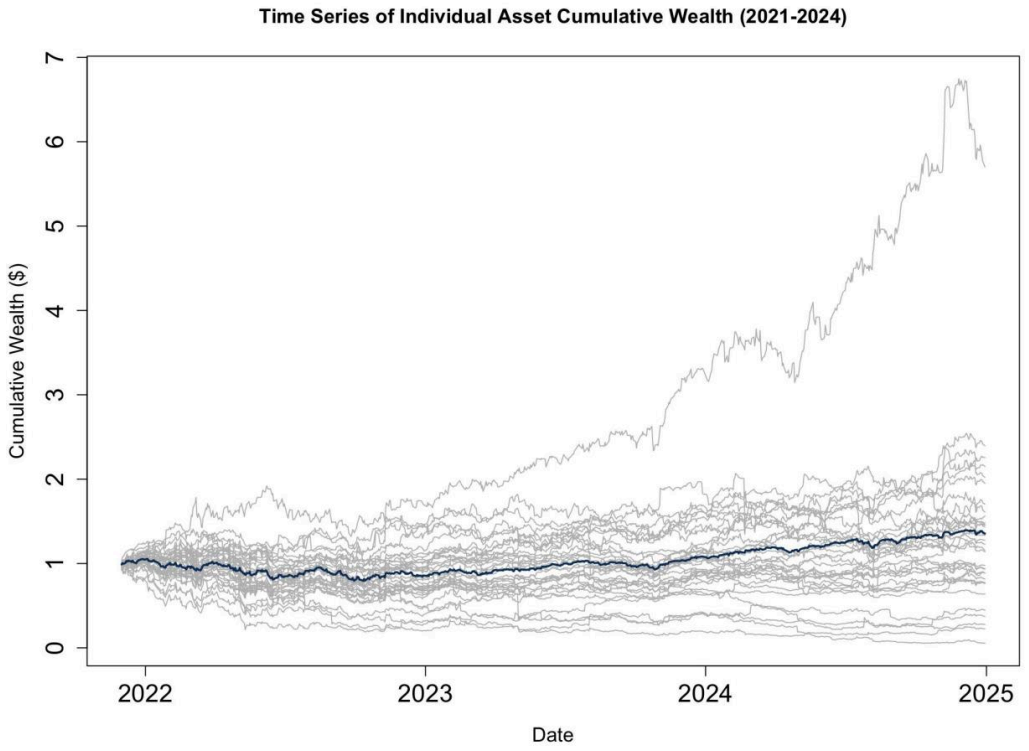


Figure 2: Cumulative wealth of individual assets

In contrast, Figure 2, built by compounding daily returns, clearly traces the non-linear divergence of asset pricing during a regime change. The figure divides into two distinct structural phases.

The first is the "systematic contraction" of 2022. Driven by the Federal Reserve's aggressive rate-hiking cycle, all assets fell together with high correlation; the broad market SPY (dark blue line) dipped below 0.8 at one point, and cross-sectional dispersion was very low. From early 2023, with the arrival of the "generative-AI-driven period", cross-sectional return dispersion widened to historic levels. SPY steadily climbed to about 1.35 (a three-year cumulative return of about 35%), but the sample also produced an extreme right-tail winner whose cumulative wealth decoupled exponentially, peaking at 6.7 (a cumulative return above 570%). Meanwhile, assets on the wrong side of the transformation drifted persistently downward, ending with only 0.1 to 0.2 of their starting value, nearly wiping out principal. This shift from "falling together" to a "K-shaped divergence" shows the tail risk of ruin that static equal-weight or single-asset buy-and-hold strategies face when an industry structure is reshuffled, and provides a solid empirical motivation for introducing dynamic mean-variance optimization models in this study.

2.2 Descriptive Statistics and Distributional Properties

To understand the return distribution of each asset, this study computes descriptive statistics for every asset over the sample period. To keep the figures comparable for financial practice, the units and annualization of each metric follow these rules:

1. **Non-annualized metrics (shown as raw decimals):** the minimum, quartiles, median and maximum. Keeping the daily extremes objectively shows the real single-day gains and losses that occurred during the sample, preserving their value as a historical stress test.

2. **Annualized metrics:** the mean and variance are annualized by multiplying by 252 trading days, and the standard deviation is annualized by the square-root-of-time rule.
3. **Moment and dependence metrics:** skewness and kurtosis are the core measures of asymmetric risk and fat tails; following the moment-scaling theorem in the course (Lecture 4), they are divided by the corresponding annualization factor to keep cross-frequency comparisons mathematically consistent. The first-order autocorrelation coefficient, ACF(1), tests whether the return series shows short-term serial dependence.

The table below summarizes the descriptive statistics of the representative assets over the three-year AI-shock sample:

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	Var	Std.	Skew	Kurt	ACF1
CHGG	−0.484	−0.024	−0.003	−0.003	0.018	0.220	0.547	0.739	−0.108	0.080	−0.112
FVRR	−0.256	−0.026	−0.005	−0.001	0.022	0.193	0.498	0.706	0.026	0.013	−0.019
UPWK	−0.180	−0.022	−0.003	0.000	0.018	0.442	0.438	0.662	0.129	0.077	−0.010
COUR	−0.167	−0.020	−0.001	−0.001	0.015	0.447	0.337	0.508	0.150	0.124	−0.078
DLB	−0.111	−0.008	0.000	0.000	0.008	0.156	0.064	0.253	0.039	0.077	0.021
PSO	−0.146	−0.007	0.001	0.001	0.009	0.177	0.088	0.297	0.100	0.093	−0.150
LOPE	−0.107	−0.008	0.001	0.001	0.009	0.149	0.078	0.279	0.107	0.068	0.076
DIS	−0.132	−0.010	0.000	0.000	0.010	0.115	0.091	0.301	−0.017	0.024	0.028
NTDOY	−0.100	−0.009	0.001	0.001	0.009	0.072	0.063	0.251	−0.010	0.012	−0.053
NYT	−0.115	−0.008	0.000	0.000	0.009	0.120	0.092	0.303	−0.001	0.031	0.011
SONY	−0.072	−0.010	0.000	0.000	0.010	0.097	0.083	0.288	0.012	0.011	0.017
WMG	−0.096	−0.011	0.000	0.000	0.012	0.152	0.133	0.365	0.015	0.020	0.048
CMCSA	−0.095	−0.008	0.000	0.000	0.009	0.103	0.069	0.263	−0.025	0.023	0.033
CRWD	−0.148	−0.016	0.002	0.001	0.019	0.130	0.278	0.527	−0.013	0.009	0.042
PANW	−0.284	−0.012	0.002	0.001	0.015	0.148	0.190	0.436	−0.061	0.069	0.017
OKTA	−0.337	−0.018	0.000	−0.001	0.018	0.265	0.365	0.604	−0.024	0.054	−0.001
EFX	−0.089	−0.011	0.001	0.000	0.012	0.137	0.105	0.324	0.002	0.015	0.025
FICO	−0.090	−0.009	0.003	0.003	0.014	0.311	0.139	0.373	0.192	0.164	0.022
FTNT	−0.251	−0.012	0.001	0.001	0.015	0.253	0.209	0.458	−0.031	0.071	−0.048
LYV	−0.123	−0.010	0.001	0.001	0.012	0.150	0.125	0.354	−0.004	0.023	−0.042
LVMUY	−0.079	−0.012	0.000	0.000	0.011	0.110	0.106	0.326	0.030	0.010	0.043
ABNB	−0.134	−0.016	0.000	0.000	0.016	0.134	0.221	0.471	−0.014	0.009	−0.008
MAR	−0.070	−0.009	0.001	0.001	0.011	0.075	0.083	0.288	−0.003	0.007	0.046
SBUX	−0.159	−0.009	−0.001	0.000	0.009	0.245	0.105	0.324	0.122	0.131	−0.019
LVS	−0.119	−0.012	0.000	0.001	0.013	0.151	0.170	0.412	0.026	0.021	0.001
DAL	−0.128	−0.013	0.001	0.001	0.015	0.093	0.140	0.375	−0.003	0.009	0.021
HCA	−0.218	−0.009	0.001	0.001	0.011	0.114	0.093	0.305	−0.113	0.095	−0.003
BKNG	−0.102	−0.009	0.002	0.001	0.013	0.079	0.106	0.326	−0.027	0.011	0.030

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	Var	Std.	Skew	Kurt	ACF1
SPY	−0.044	−0.006	0.001	0.001	0.007	0.055	0.031	0.175	−0.010	0.007	0.021
QQQ	−0.055	−0.008	0.001	0.001	0.010	0.074	0.056	0.237	−0.007	0.005	−0.003
GLD	−0.036	−0.005	0.001	0.001	0.006	0.032	0.021	0.145	−0.004	0.004	−0.027
TLT	−0.034	−0.008	0.000	−0.001	0.007	0.039	0.032	0.178	0.006	0.000	−0.059
USO	−0.117	−0.012	0.001	0.001	0.015	0.079	0.116	0.341	−0.025	0.007	0.048
DBA	−0.049	−0.005	0.001	0.001	0.007	0.030	0.022	0.147	−0.017	0.006	0.024
VNQ	−0.050	−0.007	0.000	0.000	0.008	0.073	0.042	0.206	0.007	0.009	0.031
EWW	−0.107	−0.008	0.001	0.000	0.009	0.044	0.051	0.227	−0.037	0.016	0.022
XLU	−0.047	−0.007	0.001	0.000	0.007	0.047	0.033	0.182	−0.011	0.005	0.012

Table 1: Descriptive statistics of the assets over the three-year period

2.3 Tail Risk and Autocorrelation of Individual Assets

2.3.1 The volatility surge among "AI casualties"

The data show that the online education platform Chegg (CHGG) and the outsourcing platform Fiverr (FVRR), both classified as "AI-substitution casualties", are markedly volatile. CHGG's annualized volatility reaches 73.9% and FVRR's 70.6%, both about four times that of the broad-market SPY. From an asset-pricing perspective, this abnormal volatility reflects sharp disagreement in the market over these companies' going-concern prospects and future cash flows in the face of a disruptive technology. When an industry undergoes structural change, buyers and sellers struggle to reach a pricing consensus, and prices swing in jumps. This shows that passively holding such assets exposes an investor to very high volatility risk.

2.3.2 Left-tail risk

Traditional portfolio theory often assumes normally distributed returns, but the empirical results show that most assets exhibit significant asymmetry and fat tails. Taking CHGG as an example, its annualized skewness is −0.108, indicating a left-skewed distribution in which extreme losses are more likely and larger than extreme gains. Combined with its maximum single-day loss of −48.41%, this shows that the asset carries left-tail risk of large single-day drawdowns. This combination of high volatility and negative skewness is a major challenge for optimization models: without a short-sale constraint, a model searching for minimum variance can easily assign abnormal weights because of estimation errors in the historical covariance, amplifying model risk.

2.3.3 Pricing stability of "physical moats" and "traditional safe havens"

Compared with highly volatile tech start-ups, assets with physical-service characteristics and traditional commodities show better pricing stability. For example, Marriott International (MAR) kept an annualized standard deviation of 28.78% over the three years, showing that the essential demand of the physical-experience economy offers some defensive strength. Meanwhile, the safe-haven asset gold (GLD) had an annualized standard deviation of only 14.5%, with annualized skewness (−0.004) and kurtosis (0.004) close to zero, approximating a normal distribution. In addition, most assets have first-order autocorrelations ACF(1) roughly between −0.02 and −0.11, showing only slight single-day reversals and no significant long-

term trend dependence, consistent with the micro-level features of the weak-form efficient market hypothesis.

2.3.4 Summary

The results of this section show that under the AI technology shock, the return distributions and risk characteristics of the sample assets diverged significantly across the cross-section. If an investor simply adopts a traditional equal-weight strategy without accounting for individual assets' high volatility and left-tail risk, the overall portfolio will bear excessive downside risk. These differences among individual assets establish the empirical motivation for introducing Markowitz optimization for dynamic asset allocation in Chapters 3 and 4.

2.4 Correlation Matrix and Diversification Potential

According to Markowitz's Modern Portfolio Theory, the key to lowering portfolio volatility is choosing assets with low mutual correlation. This section examines the correlation matrix of the three-year sample to assess how these 37 assets of different natures moved together during the AI wave.

	CHGG	FVRR	UPWK	COUR	DLB	PSO	LOPE	DIS	NTDOY
CHGG	1	0.368	0.298	0.368	0.200	0.241	0.247	0.228	0.120
FVRR	0.368	1	0.677	0.452	0.359	0.185	0.216	0.400	0.204
UPWK	0.298	0.677	1	0.435	0.343	0.173	0.236	0.377	0.221
COUR	0.368	0.452	0.435	1	0.307	0.188	0.257	0.347	0.198
DLB	0.200	0.359	0.343	0.307	1	0.150	0.207	0.389	0.213
PSO	0.241	0.185	0.173	0.188	0.150	1	0.159	0.219	0.154
LOPE	0.247	0.216	0.236	0.257	0.207	0.159	1	0.211	0.109
DIS	0.228	0.400	0.377	0.347	0.389	0.219	0.211	1	0.270
NTDOY	0.120	0.204	0.221	0.198	0.213	0.154	0.109	0.270	1
NYT	0.290	0.387	0.340	0.311	0.323	0.166	0.254	0.352	0.241
SONY	0.215	0.374	0.321	0.322	0.394	0.202	0.186	0.341	0.316
WMG	0.191	0.395	0.364	0.281	0.302	0.182	0.169	0.324	0.186
CMCSA	0.196	0.320	0.277	0.308	0.313	0.214	0.260	0.407	0.161
CRWD	0.182	0.486	0.434	0.355	0.311	0.133	0.164	0.395	0.207
PANW	0.130	0.368	0.275	0.287	0.260	0.141	0.161	0.326	0.157
OKTA	0.227	0.474	0.415	0.352	0.296	0.089	0.228	0.368	0.198
EFX	0.255	0.466	0.402	0.376	0.386	0.220	0.240	0.401	0.223
FICO	0.134	0.377	0.375	0.267	0.339	0.158	0.236	0.346	0.245
FTNT	0.152	0.354	0.319	0.280	0.363	0.117	0.097	0.298	0.213
LYV	0.277	0.418	0.398	0.339	0.345	0.200	0.282	0.452	0.237
LVMUY	0.193	0.338	0.314	0.317	0.388	0.275	0.177	0.422	0.347
ABNB	0.261	0.513	0.432	0.436	0.377	0.202	0.242	0.522	0.229
MAR	0.186	0.358	0.308	0.364	0.360	0.211	0.306	0.496	0.247

	CHGG	FVRR	UPWK	COUR	DLB	PSO	LOPE	DIS	NTDOY
SBUX	0.185	0.378	0.318	0.285	0.360	0.066	0.198	0.388	0.182
LVS	0.244	0.365	0.347	0.321	0.334	0.103	0.260	0.393	0.250
DAL	0.199	0.393	0.325	0.346	0.337	0.286	0.283	0.481	0.243
HCA	0.139	0.238	0.207	0.189	0.261	0.205	0.166	0.275	0.133
BKNG	0.190	0.349	0.336	0.328	0.305	0.219	0.275	0.477	0.260
SPY	0.287	0.534	0.471	0.461	0.526	0.312	0.363	0.600	0.351
QQQ	0.270	0.580	0.513	0.476	0.501	0.270	0.320	0.557	0.334
GLD	0.068	0.089	0.041	0.097	0.071	0.058	0.062	0.077	0.094
TLT	0.052	0.115	0.113	0.119	0.085	0.051	0.046	0.021	0.073
USO	0.031	0.021	0.032	0.047	−0.007	−0.019	0.025	0.100	0.041
DBA	0.026	0.011	−0.004	0.050	0.065	−0.003	0.018	0.047	0.026
VNQ	0.244	0.429	0.380	0.361	0.450	0.268	0.280	0.454	0.264
EWW	0.222	0.302	0.279	0.271	0.289	0.217	0.235	0.331	0.277
XLU	0.150	0.177	0.107	0.136	0.270	0.190	0.184	0.279	0.128

Table 2: Correlation matrix, part 1

	NYT	SONY	WMG	CMCSA	CRWD	PANW	OKTA	EFX	FICO
CHGG	0.290	0.215	0.191	0.196	0.182	0.130	0.227	0.255	0.134
FVRR	0.387	0.374	0.395	0.320	0.486	0.368	0.474	0.466	0.377
UPWK	0.340	0.321	0.364	0.277	0.434	0.275	0.415	0.402	0.375
COUR	0.311	0.322	0.281	0.308	0.355	0.287	0.352	0.376	0.267
DLB	0.323	0.394	0.302	0.313	0.311	0.260	0.296	0.386	0.339
PSO	0.166	0.202	0.182	0.214	0.133	0.141	0.089	0.220	0.158
LOPE	0.254	0.186	0.169	0.260	0.164	0.161	0.228	0.240	0.236
DIS	0.352	0.341	0.324	0.407	0.395	0.326	0.368	0.401	0.346
NTDOY	0.241	0.316	0.186	0.161	0.207	0.157	0.198	0.223	0.245
NYT	1	0.325	0.337	0.346	0.300	0.279	0.316	0.366	0.311
SONY	0.325	1	0.366	0.325	0.342	0.265	0.316	0.410	0.368
WMG	0.337	0.366	1	0.310	0.321	0.269	0.325	0.321	0.256
CMCSA	0.346	0.325	0.310	1	0.196	0.203	0.246	0.305	0.275
CRWD	0.300	0.342	0.321	0.196	1	0.628	0.575	0.400	0.429
PANW	0.279	0.265	0.269	0.203	0.628	1	0.442	0.344	0.385
OKTA	0.316	0.316	0.325	0.246	0.575	0.442	1	0.392	0.386
EFX	0.366	0.410	0.321	0.305	0.400	0.344	0.392	1	0.571
FICO	0.311	0.368	0.256	0.275	0.429	0.385	0.386	0.571	1
FTNT	0.316	0.348	0.309	0.238	0.524	0.573	0.419	0.379	0.404
LYV	0.408	0.357	0.391	0.366	0.457	0.355	0.443	0.417	0.385

	NYT	SONY	WMG	CMCSA	CRWD	PANW	OKTA	EFX	FICO
LVMUY	0.283	0.387	0.327	0.289	0.269	0.231	0.275	0.428	0.334
ABNB	0.353	0.423	0.401	0.322	0.450	0.345	0.447	0.401	0.366
MAR	0.352	0.415	0.316	0.399	0.399	0.326	0.371	0.449	0.421
SBUX	0.305	0.365	0.279	0.320	0.256	0.211	0.270	0.394	0.318
LVS	0.315	0.337	0.229	0.243	0.292	0.213	0.324	0.311	0.247
DAL	0.335	0.344	0.329	0.356	0.339	0.272	0.316	0.368	0.319
HCA	0.234	0.302	0.196	0.307	0.233	0.190	0.186	0.420	0.312
BKNG	0.298	0.394	0.336	0.341	0.376	0.283	0.344	0.356	0.357
SPY	0.495	0.577	0.450	0.497	0.566	0.502	0.506	0.649	0.576
QQQ	0.471	0.574	0.453	0.444	0.626	0.545	0.553	0.616	0.569
GLD	0.071	0.169	0.054	0.046	0.066	0.005	0.089	0.152	0.116
TLT	0.095	0.112	0.065	0.083	0.041	0.037	0.125	0.257	0.182
USO	0.079	0.064	0.031	0.045	0.073	0.036	0.045	0.012	0.010
DBA	0.050	0.037	0.002	0.020	0.041	0.035	0.007	0.043	0.034
VNQ	0.405	0.456	0.318	0.402	0.335	0.264	0.380	0.648	0.471
EWW	0.303	0.342	0.258	0.309	0.239	0.205	0.259	0.370	0.266
XLU	0.239	0.237	0.134	0.307	0.108	0.116	0.137	0.398	0.267

Table 3: Correlation matrix, part 2

	FTNT	LYV	LVMUY	ABNB	MAR	SBUX	LVS	DAL	HCA
CHGG	0.152	0.277	0.193	0.261	0.186	0.185	0.244	0.199	0.139
FVRR	0.354	0.418	0.338	0.513	0.358	0.378	0.365	0.393	0.238
UPWK	0.319	0.398	0.314	0.432	0.308	0.318	0.347	0.325	0.207
COUR	0.280	0.339	0.317	0.436	0.364	0.285	0.321	0.346	0.189
DLB	0.363	0.345	0.388	0.377	0.360	0.360	0.334	0.337	0.261
PSO	0.117	0.200	0.275	0.202	0.211	0.066	0.103	0.286	0.205
LOPE	0.097	0.282	0.177	0.242	0.306	0.198	0.260	0.283	0.166
DIS	0.298	0.452	0.422	0.522	0.496	0.388	0.393	0.481	0.275
NTDOY	0.213	0.237	0.347	0.229	0.247	0.182	0.250	0.243	0.133
NYT	0.316	0.408	0.283	0.353	0.352	0.305	0.315	0.335	0.234
SONY	0.348	0.357	0.387	0.423	0.415	0.365	0.337	0.344	0.302
WMG	0.309	0.391	0.327	0.401	0.316	0.279	0.229	0.329	0.196
CMCSA	0.238	0.366	0.289	0.322	0.399	0.320	0.243	0.356	0.307
CRWD	0.524	0.457	0.269	0.450	0.399	0.256	0.292	0.339	0.233
PANW	0.573	0.355	0.231	0.345	0.326	0.211	0.213	0.272	0.190
OKTA	0.419	0.443	0.275	0.447	0.371	0.270	0.324	0.316	0.186
EFX	0.379	0.417	0.428	0.401	0.449	0.394	0.311	0.368	0.420

	FTNT	LYV	LVMUY	ABNB	MAR	SBUX	LVS	DAL	HCA
FICO	0.404	0.385	0.334	0.366	0.421	0.318	0.247	0.319	0.312
FTNT	1	0.382	0.289	0.297	0.369	0.305	0.271	0.281	0.267
LYV	0.382	1	0.337	0.526	0.554	0.336	0.393	0.503	0.270
LVMUY	0.289	0.337	1	0.458	0.473	0.406	0.447	0.453	0.284
ABNB	0.297	0.526	0.458	1	0.635	0.401	0.468	0.543	0.238
MAR	0.369	0.554	0.473	0.635	1	0.408	0.484	0.645	0.333
SBUX	0.305	0.336	0.406	0.401	0.408	1	0.433	0.383	0.285
LVS	0.271	0.393	0.447	0.468	0.484	0.433	1	0.427	0.217
DAL	0.281	0.503	0.453	0.543	0.645	0.383	0.427	1	0.308
HCA	0.267	0.270	0.284	0.238	0.333	0.285	0.217	0.308	1
BKNG	0.256	0.515	0.458	0.618	0.671	0.422	0.482	0.590	0.298
SPY	0.535	0.585	0.587	0.622	0.658	0.536	0.473	0.596	0.487
QQQ	0.545	0.583	0.551	0.647	0.611	0.506	0.463	0.539	0.391
GLD	0.033	0.010	0.177	0.078	0.064	0.069	0.060	−0.011	0.150
TLT	0.052	0.024	0.088	0.009	−0.005	0.089	−0.037	0.023	0.144
USO	0.066	0.051	0.044	0.104	0.074	0.041	0.103	−0.035	0.075
DBA	0.042	0.109	0.065	0.058	0.054	0.006	−0.016	−0.004	0.062
VNQ	0.358	0.448	0.450	0.433	0.459	0.431	0.319	0.447	0.475
EWW	0.251	0.359	0.460	0.356	0.429	0.315	0.320	0.416	0.311
XLU	0.208	0.183	0.269	0.156	0.217	0.298	0.118	0.229	0.396

Table 4: Correlation matrix, part 3

	BKNG	SPY	QQQ	GLD	TLT	USO	DBA	VNQ	EWW	XLU
CHGG	0.190	0.287	0.270	0.068	0.052	0.031	0.026	0.244	0.222	0.150
FVRR	0.349	0.534	0.580	0.089	0.115	0.021	0.011	0.429	0.302	0.177
UPWK	0.336	0.471	0.513	0.041	0.113	0.032	−0.004	0.380	0.279	0.107
COUR	0.328	0.461	0.476	0.097	0.119	0.047	0.050	0.361	0.271	0.136
DLB	0.305	0.526	0.501	0.071	0.085	−0.007	0.065	0.450	0.289	0.270
PSO	0.219	0.312	0.270	0.058	0.051	−0.019	−0.003	0.268	0.217	0.190
LOPE	0.275	0.363	0.320	0.062	0.046	0.025	0.018	0.280	0.235	0.184
DIS	0.477	0.600	0.557	0.077	0.021	0.100	0.047	0.454	0.331	0.279
NTDOY	0.260	0.351	0.334	0.094	0.073	0.041	0.026	0.264	0.277	0.128
NYT	0.298	0.495	0.471	0.071	0.095	0.079	0.050	0.405	0.303	0.239
SONY	0.394	0.577	0.574	0.169	0.112	0.064	0.037	0.456	0.342	0.237
WMG	0.336	0.450	0.453	0.054	0.065	0.031	0.002	0.318	0.258	0.134
CMCSA	0.341	0.497	0.444	0.046	0.083	0.045	0.020	0.402	0.309	0.307
CRWD	0.376	0.566	0.626	0.066	0.041	0.073	0.041	0.335	0.239	0.108

	BKNG	SPY	QQQ	GLD	TLT	USO	DBA	VNQ	EWW	XLU
PANW	0.283	0.502	0.545	0.005	0.037	0.036	0.035	0.264	0.205	0.116
OKTA	0.344	0.506	0.553	0.089	0.125	0.045	0.007	0.380	0.259	0.137
EFX	0.356	0.649	0.616	0.152	0.257	0.012	0.043	0.648	0.370	0.398
FICO	0.357	0.576	0.569	0.116	0.182	0.010	0.034	0.471	0.266	0.267
FTNT	0.256	0.535	0.545	0.033	0.052	0.066	0.042	0.358	0.251	0.208
LYV	0.515	0.585	0.583	0.010	0.024	0.051	0.109	0.448	0.359	0.183
LVMUY	0.458	0.587	0.551	0.177	0.088	0.044	0.065	0.450	0.460	0.269
ABNB	0.618	0.622	0.647	0.078	0.009	0.104	0.058	0.433	0.356	0.156
MAR	0.671	0.658	0.611	0.064	−0.005	0.074	0.054	0.459	0.429	0.217
SBUX	0.422	0.536	0.506	0.069	0.089	0.041	0.006	0.431	0.315	0.298
LVS	0.482	0.473	0.463	0.060	−0.037	0.103	−0.016	0.319	0.320	0.118
DAL	0.590	0.596	0.539	−0.011	0.023	−0.035	−0.004	0.447	0.416	0.229
HCA	0.298	0.487	0.391	0.150	0.144	0.075	0.062	0.475	0.311	0.396
BKNG	1	0.613	0.591	0.050	−0.034	0.065	0.053	0.392	0.429	0.179
SPY	0.613	1	0.952	0.161	0.080	0.127	0.108	0.718	0.542	0.482
QQQ	0.591	0.952	1	0.132	0.075	0.064	0.084	0.594	0.477	0.323
GLD	0.050	0.161	0.132	1	0.324	0.263	0.169	0.230	0.221	0.263
TLT	−0.034	0.080	0.075	0.324	1	−0.133	−0.046	0.295	0.155	0.262
USO	0.065	0.127	0.064	0.263	−0.133	1	0.296	0.056	0.157	0.068
DBA	0.053	0.108	0.084	0.169	−0.046	0.296	1	0.092	0.114	0.035
VNQ	0.392	0.718	0.594	0.230	0.295	0.056	0.092	1	0.456	0.676
EWW	0.429	0.542	0.477	0.221	0.155	0.157	0.114	0.456	1	0.323
XLU	0.179	0.482	0.323	0.263	0.262	0.068	0.035	0.676	0.323	1

Table 5: Correlation matrix, part 4

Several important patterns emerge from the correlation matrix:

1. **Tech benchmarks have decoupled from "AI casualties".** The correlation between the traditional tech benchmark (QQQ) and assets regarded as high-risk AI substitutes (such as CHGG and FVRR) is lower than expected (for example, QQQ and CHGG have a correlation of only 0.27). This shows that the market already treats these companies as decoupled from the broader tech rally; their risk profiles in the face of technological progress have clearly diverged.
2. **The protective role of traditional safe havens.** The correlations of gold (GLD) and long-term Treasuries (TLT) with the broad equity index (SPY) stay at very low levels (GLD with SPY at 0.161; TLT with SPY at 0.080), and TLT is even negatively correlated with crude oil (USO, −0.133). Adding such low- or negatively correlated assets to a portfolio provides an effective cushion when equity markets fall.
3. **The independence of the physical-experience economy.** Industries that emphasize in-person contact and experience, such as luxury goods (LVMUY) and live entertainment (LYV), have relatively low

correlations with the AI-shock group (for example, LVMUY and CHGG at only 0.193). This means that sectors with a "physical moat" are less affected by AI-driven change and have independent pricing power and resilience.

Overall, the 37 assets selected in this study are highly heterogeneous by industry. The many low- and negatively correlated pairs in the matrix provide a strong basis for risk diversification as the next chapter uses quantitative models to find optimal allocation weights.

3. Portfolio Strategies and Out-of-Sample Backtesting Framework

Having examined the distributional features and correlation structure of individual assets in the previous chapter, this chapter applies Markowitz's Modern Portfolio Theory, through mean-variance analysis, to build systematic quantitative portfolios. To objectively assess the effect of different constraints on model performance, the study implements seven strategies with different financial meanings, divided into two categories, "passive benchmarks" and "active optimization", for cross-comparison of performance.

3.1 Portfolio Models

3.1.1 Passive Benchmarks

1. **1/N Equal Weight Portfolio.** This strategy allocates capital equally across the 37 assets on every trading day. It implies that the investor has no prior information about expected returns or covariances (naive diversification). Although it requires no parameter estimation, it is easily dragged down by highly volatile assets; this study uses it as the baseline for judging whether active strategies add excess value.
2. **Buy and Hold (B&H).** After investing capital at equal weights at the start, this strategy performs no further rebalancing. Asset weights drift naturally with daily price changes, and the weights of strong performers expand automatically. It represents the actual outcome for a passive investor facing a structural market change.

3.1.2 Active Optimization Portfolios

This study uses the R package quadprog to solve quadratic programs for the optimal asset weights under given constraints.

3. **Global Minimum Variance Portfolio (GMVP).** Because historical mean returns are estimated with large error, this strategy discards expected-return parameters and relies only on the historical covariance matrix to minimize portfolio variance. Short sales are allowed, and risk is offset through long-short hedging.
4. **No-Shortsale GMVP (NS-GMVP).** Because institutional investors are often subject to short-sale restrictions in practice, this model adds a non-negative-weight constraint to the GMVP. The constraint helps suppress extreme, overly concentrated weights caused by estimation error, improving the portfolio's robustness.
5. **MVP with Target Return.** With short sales allowed, the model minimizes variance subject to an equality constraint that the portfolio meets a specific target expected return, reflecting an investor's baseline requirement for capital growth.
6. **No-Shortsale MVP (NS-MVP).** The practical, constrained version of the MVP. With short sales disallowed, the model can only search a narrower feasible space for an allocation across high-volatility tech stocks and low-volatility defensive assets that meets the target return.
7. **Tangency Portfolio.** Incorporating a risk-free asset, the model seeks the point on the capital allocation line that maximizes the Sharpe ratio. Because the strategy depends heavily on historical expected returns as inputs, its weights tend to become extreme; this study uses it as an indicator of model risk.

3.2 Model Parameters and Constraints

To ensure that the quantitative models reflect the macroeconomic environment and trading practice, the core parameters and constraints are set as follows:

1. **Target return.** Set at 10% per year, with reference to the long-run historical average equity premium of the S&P 500, as a reasonable baseline of capital growth the optimization model must achieve.
2. **Risk-free rate.** Set at 2.5% per year. Because the sample spans both prolonged quantitative easing (zero rates) and the recent aggressive rate-hiking cycle, the ten-year historical average of the 10-year US Treasury yield is used as the discount benchmark, approximating the real opportunity cost of long-term investment.
3. **Transaction costs.** Set at 0.35% (35 basis points, i.e., 0.0035) of traded value. In a dynamic backtest, frequent buying and selling incurs slippage, commissions and other frictions. The study strictly includes these costs in net returns to avoid "paper returns" that could not be achieved in practice.
4. **Weight constraints.** When running the solve.QP solver, every strategy is subject to the budget constraint that weights sum to 100%. The NS-GMVP and NS-MVP models additionally include an N×N identity matrix to impose non-negative weights.

3.3 Out-of-Sample Rolling-Window Design

To approximate real investment conditions and avoid the look-ahead bias that arises from backtesting with future data, this study uses an out-of-sample rolling-window framework to test the real performance of each portfolio strategy.

The dynamic rolling loop proceeds as follows:

1. **Estimation window.** The rolling estimation window is fixed at k = 252 days (one year of trading days). At any time point t in the loop, the model uses only the known daily returns of the previous 252 days to estimate the covariance matrix and mean returns. For example, the first matrix uses historical data from 1 December 2021 to 30 November 2022, fully eliminating look-ahead bias.
2. **Out-of-sample testing.** The out-of-sample testing period runs from 1 December 2022 to 31 December 2024 (h = 522 days). The model multiplies the optimal weights solved on day t from known information by the actual, previously unknown returns of day t to compute each day's out-of-sample profit or loss. As the window rolls forward one day at a time, this produces a complete out-of-sample net return series of 522 days.
3. **Weight drift and turnover adjustment.** Because assets rise and fall by different amounts each day, starting weights drift. When computing daily turnover, the study calculates the absolute difference between "today's target weights" and "yesterday's drifted weights". Only after deducting the transaction costs implied by turnover is the final daily net return computed, ensuring the results are practically implementable.

4. Empirical Results and Advanced Risk Metrics

This chapter presents the empirical results of the out-of-sample (OoS) rolling backtest. By applying the seven portfolio strategies to unseen market data and deducting transaction costs, it evaluates how each model actually responds to the AI-wave shock.

4.1 Out-of-Sample Cumulative Wealth and Returns

When examining out-of-sample performance, time-series charts give the most intuitive view of each strategy's characteristics.

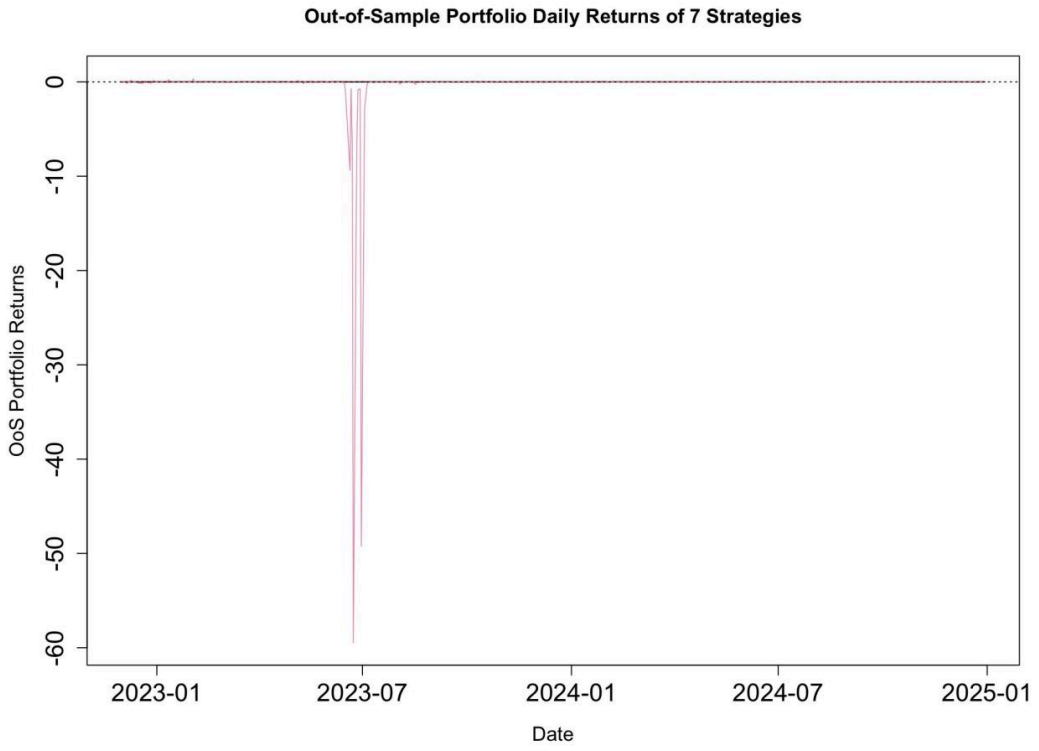


Figure 3: Out-of-sample (OoS) daily net returns of the seven portfolio strategies

As the OoS daily net-return chart shows, the passive strategies (1/N and B&H) fluctuate in a fairly consistent range; however, the active optimization strategies that allow short sales (such as Tangency and MVP) show extremely violent swings on specific trading days. This confirms that when the market suffers an unexpected structural shock, models that rely heavily on historical expected returns easily produce extreme weights, leading to violent jumps in daily profit and loss.

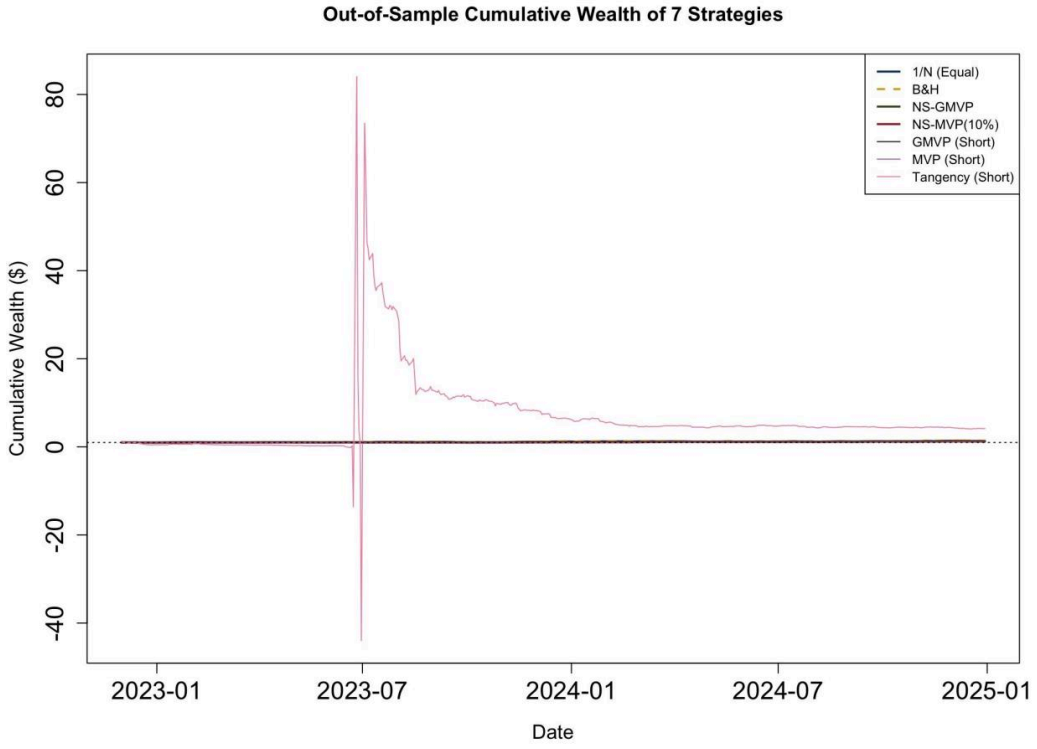


Figure 4: OoS cumulative wealth (full view)

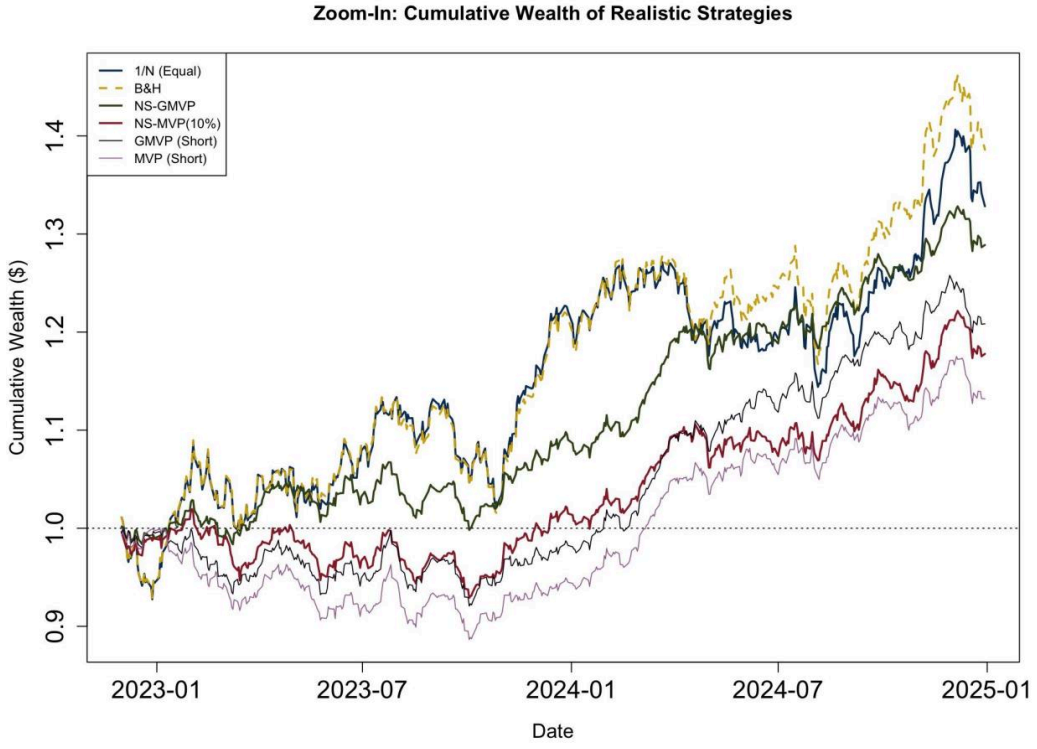


Figure 5: OoS cumulative wealth (zoomed in)

Looking further at the OoS cumulative wealth chart (initial capital set at $1), we find that the Tangency strategy without a short-sale constraint suffered severe estimation error and short-squeeze risk, and its

cumulative wealth path broke down. To compare the other, practically feasible strategies more clearly, the zoomed-in chart that excludes the Tangency outliers shows the following:

1. **Passive strategies bear high volatility.** Although the 1/N equal-weight strategy and the buy-and-hold (B&H) strategy ended the period with positive cumulative wealth, their climb was accompanied by large swings. This shows that when the market experiences a structural shock, blindly spreading capital to casualty assets exposes the portfolio to unnecessary market volatility.
2. **No-short-sale strategies show defensive strength.** The NS-GMVP strategy, with its non-negative weight constraint, climbed very smoothly. In contrast, the unconstrained Tangency strategy broke down destructively because of severe estimation error and short-squeeze effects. This demonstrates the practical protective value of restricting short sales in extreme market conditions.

4.2 Core Performance Evaluation

To quantify strategy performance, this section compares each strategy's annualized return, annualized volatility and Sharpe ratio.

	1/N	B&H	GMVP	NS-GMVP	MVP (10%)	NS-MVP (10%)	Tangency
Ann_Ret	0.148	0.168	0.095	0.126	0.063	0.082	−68.406
Ann_Vol	0.146	0.145	0.080	0.081	0.082	0.083	54.144
Sharpe	0.841	0.986	0.871	1.249	0.464	0.693	−1.264
VaR_05	−0.015	−0.016	−0.008	−0.008	−0.009	−0.009	−0.091
ES_05	−0.020	−0.020	−0.010	−0.011	−0.011	−0.011	−5.201
LPSD	0.097	0.097	0.050	0.052	0.053	0.053	71.155
Turnover	0.012	0.000	0.086	0.029	0.117	0.069	34.198
HHI	0.027	0.029	0.144	0.213	0.128	0.208	0.107
SLR	0.000	0.000	0.459	0.000	0.455	0.000	0.801
MDD	0.100	0.104	0.079	0.065	0.114	0.089	13.437

Table 6: OoS performance summary

The summary table yields the following financial implications:

1. **The risk-return trade-off.** Among all regular strategies, the B&H (buy-and-hold) strategy, benefiting from the broad bull market, achieved the highest annualized return of 16.8%, but its annualized volatility was also high at 14.5%. By contrast, NS-GMVP (the no-short-sale minimum variance portfolio) succeeded in pushing annualized volatility down to the lowest level, 8.1% (very close to GMVP's 8.0%), fully achieving its defensive goal of absorbing shocks.
2. **Sharpe ratio comparison.** After deducting the 2.5% risk-free rate, NS-GMVP's Sharpe ratio reached 1.249, not only the best of all strategies but also well above the 1/N benchmark (0.841) and B&H (0.986). This shows that even without any return forecasting, simply minimizing risk through the historical covariance matrix with a short-sale constraint still delivered the best risk-adjusted return during the AI shock.

4.3 Downside Risk and Extreme Drawdowns

Traditional standard deviation treats upside and downside moves as equal risk, but practitioners care more about downside risk. This study further introduces VaR (5%), expected shortfall (ES, 5%), lower partial standard deviation (LPSD) and maximum drawdown (MDD).

1. **Extreme tail risk (VaR and ES).** At the 95% confidence level, the 1/N strategy's worst-case expected single-day loss (ES) is −2.0%, whereas the NS-GMVP strategy, which diversifies risk through an optimization algorithm, cuts the extreme single-day loss substantially to −1.1%. This shows that optimization models can effectively withstand left-tail shocks.
2. **Maximum drawdown (MDD).** MDD measures the largest loss of principal an investor could face by entering at the worst possible time. The results show that the 1/N strategy's MDD reached 10.0%, and the MVP strategy with a target-return constraint reached as high as 11.4%; by contrast, NS-GMVP's MDD was only 6.5%, the strongest downside protection in this study.

4.4 Implementation Constraints: Turnover, Concentration (HHI) and Short-to-Long Ratio (SLR)

1. **Frictional cost of turnover.** After deducting transaction costs, we find that the unconstrained active strategies (such as MVP and Tangency) had average daily turnover as high as 0.117 and 34.198 respectively, meaning the models bought and sold assets frequently and in large amounts, so potential profits were heavily eroded or even wiped out by transaction costs. NS-GMVP's turnover was only 0.029, achieving the best practical balance.
2. **Concentration (HHI).** The 1/N strategy necessarily has a very low HHI of 0.027 (complete diversification). However, the HHI of the optimization models (especially NS-GMVP) rises to 0.213. This reflects that during the rolling process the model actively screens out high-risk AI-casualty assets and concentrates capital in low-correlation or low-volatility defensive assets.
3. **Short-to-long ratio (SLR).** The passive strategies and the NS strategies all have an SLR of 0. However, the unconstrained GMVP strategy has an SLR as high as 0.459 (short positions equal to 45.9% of long positions). In real financial markets, relying heavily on short positions exposes a portfolio to high margin-call and maintenance-ratio risk, which further underscores the necessity of a no-short-sale condition.

5. Conclusion

5.1 Summary of Empirical Results

Set against the structural market change brought by the generative-AI boom from 2021 to 2024, this study selected 37 assets spanning "AI-substitution casualties", "physical moats" and "traditional safe havens" and, through a rigorous out-of-sample rolling-window framework, tested the empirical performance of Markowitz portfolio-optimization models under extreme market conditions. Based on the empirical data, the study draws three core conclusions:

First, passive diversification has blind spots in extreme markets. Although the buy-and-hold (B&H) strategy, benefiting from the broad bull market, achieved the highest annualized return of 16.8%, it, like the 1/N strategy, was exposed to high volatility above 14.5% and a maximum drawdown of 10.0%. Allocating capital blindly and evenly forces investors to bear, indiscriminately, the left-tail collapse risk of "AI-substitution casualties" (such as CHGG and FVRR) whose business models are breaking down.

Second, unconstrained optimization models easily fall into the "error maximization" trap. The results show that the short-allowed Tangency and MVP portfolios performed very poorly; the Tangency strategy in particular was wiped out by extreme turnover (34.198) and short-squeeze effects. This confirms that when markets are non-stationary, historical expected returns and covariance matrices are full of estimation error, and the algorithm mistakes high-risk assets for hedging instruments, amplifying the fragility of the overall portfolio.

Third, the "no-short-sale global minimum variance model (NS-GMVP)" showed the best practical defensive strength. The NS-GMVP strategy not only avoided enormous transaction frictions but also pushed annualized volatility down to 8.1% and held maximum drawdown to 6.5%, producing the best Sharpe ratio of all strategies (1.249). This shows that the quantitative model does have a keen sense of risk and can automatically withdraw capital from AI casualties and concentrate it in defensive assets with a "human premium" and a "physical moat".

5.2 Practical Limitations and Recommendations for Future Research

Although the NS-GMVP strategy delivered excellent risk-adjusted returns within this sample period, from the perspective of practical quantitative trading the model still has some limitations, and the following directions are proposed for future research:

First, regarding covariance estimation, this study uses the traditional sample covariance matrix with a fixed 252-day rolling window. However, financial-market volatility shows significant clustering and time variation. Future research could adopt an exponentially weighted moving average (EWMA) or a generalized autoregressive conditional heteroskedasticity (GARCH) model to give recent data greater weight and improve the accuracy of matrix forecasts.

Second, with a relatively large number of assets (N = 37), inverting the sample matrix is highly sensitive to extreme values. Future research could introduce shrinkage estimators (such as the Ledoit-Wolf method), which linearly shrink the sample covariance toward a target matrix, further reducing estimation error and improving the out-of-sample stability of optimized weights.

In conclusion, in the face of the unknown industry shocks brought by AI technology, the "human premium" and physical assets do provide a reliable safe haven for capital. However, investors cannot rely solely on passive diversification; only a quantitative optimization framework with strictly set practical constraints (such as no short sales and accounting for transaction costs) can, in this technology boom, pursue capital growth while steadily weathering the test of tail risk.